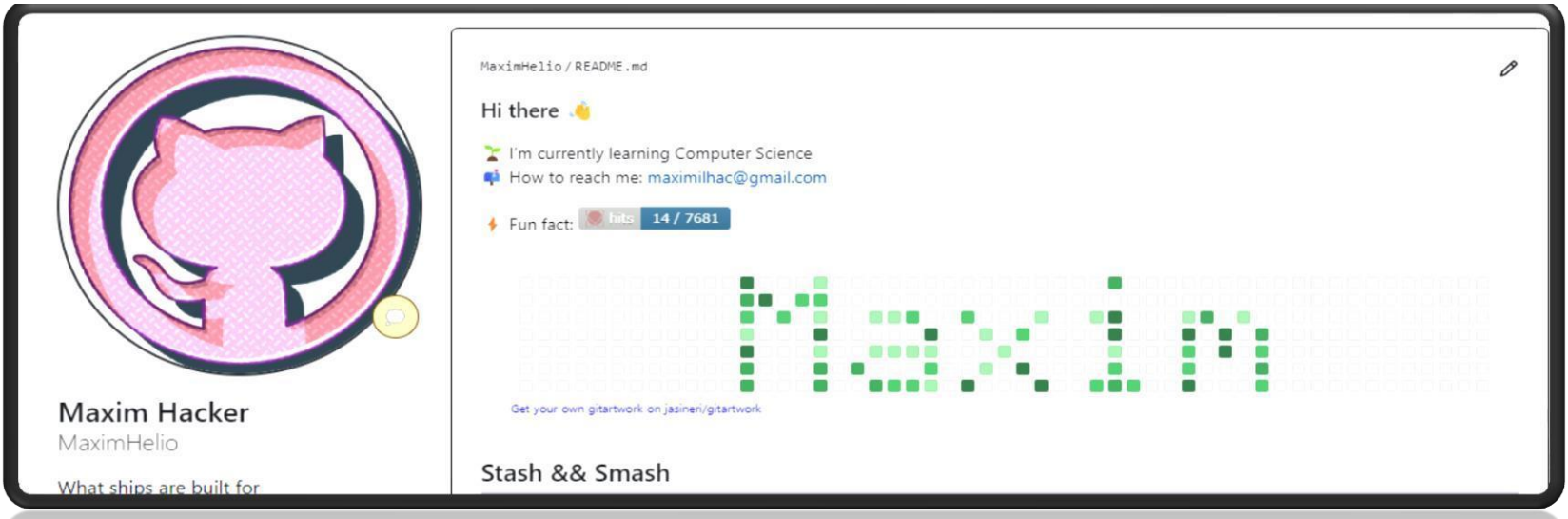
HAEJUNG KOH
PORTFOLIO



Main skills

- C++, Java, Javascript, SQL
- Mathematica, Gromacs, Gaussian 05, Origin, TensorFlow
- Spring Boot, Node.js, Nuxt.js, Three.js, Laravel, pro*C, Adobe flex, FastAPI, Django, Vue.js
- MySQL, Oracle, MongoDB, AWS
- Intellij, Eclipse, Visual Studio Code
- Git, SVN

Repository https://github.com/MaximHelio

Education

- B.Eng. Computer Engineering
- B.Sc. Chemistry and Nanoscience

Work Experience

- Samsungcard Digital Service dept. SW Engineer
- Shinhan Global dept. SW Engineer/Expatriate
- Sandslab PLC. Web development dept. SW Engineer

Certificates

- Structured Query Language Development
- Engineer Information Processing
- Microsoft Office Master 2010

Interests

- Data analysis, Automatic Program
- Simulations
- Help others

Mydata Construction Development of Samsung Card Networking Mobile Application, *Monimo*

https://www.monimo.com/

Senior SW engineer Samsung Card

Skill stack

FE: Javascript-Nuxt.js, Vue.js

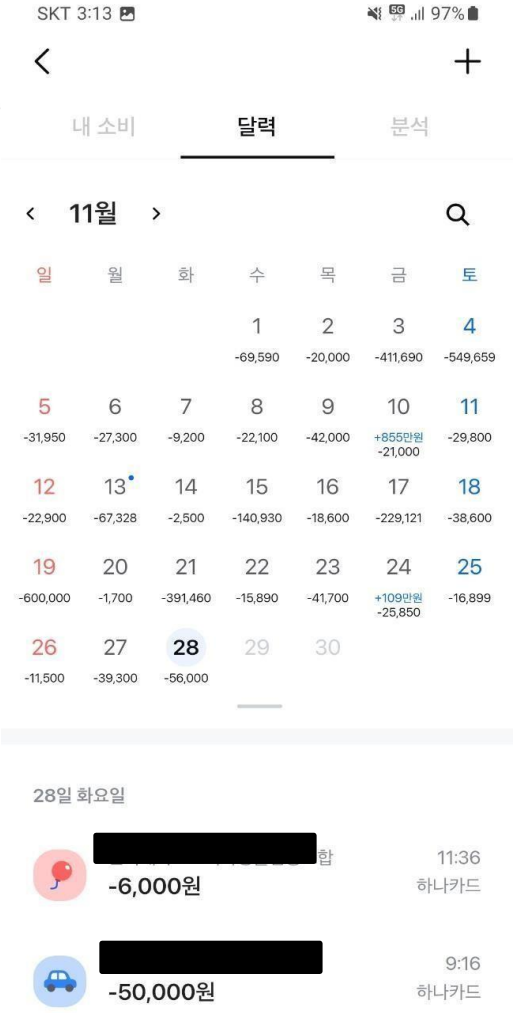
BE: Java-Spring Boot 2.x, Oracle

RNRs

FE/BE

PJT Outline

- Developed all screens and functions of Consummation, Point money, Pension, and Insurance
- Developed real-time search and batch processes for representation of consummation in the calendar
- Integrated Mydata, users' financial transactions and activity data, into applications
- Depicted the consummation graphs and applied card product recommendation algorithms according to the consummation patterns



Development of Cyber Security Web Malwares.com

Junior SW Engineer, SandsLab PLC.

PJT Outline

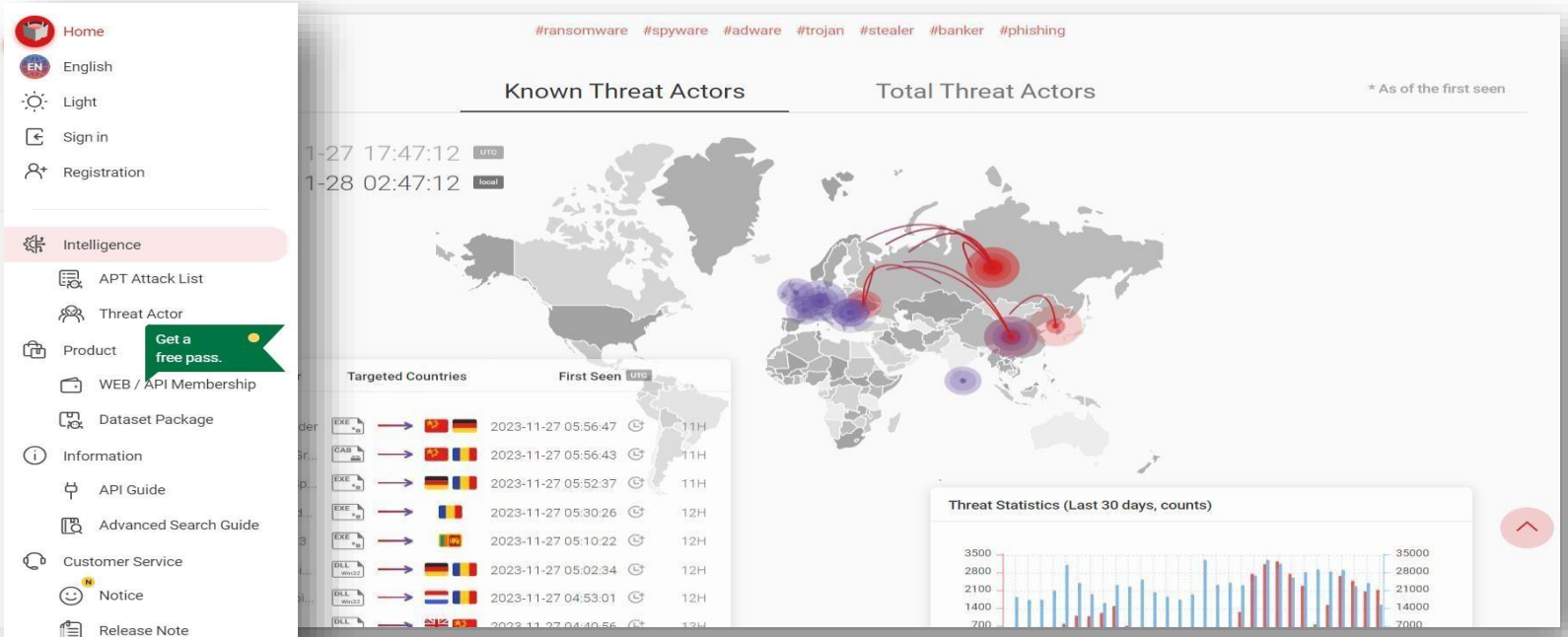
Developed Company solution homepage malwares.com
Published all Front-End user interface
Created an internal API and Private API
Developed payment system Develope
d multi-modal functions Developed a r
eal-time search system
Developed fragmentation search functions (i.e. author = haejung)

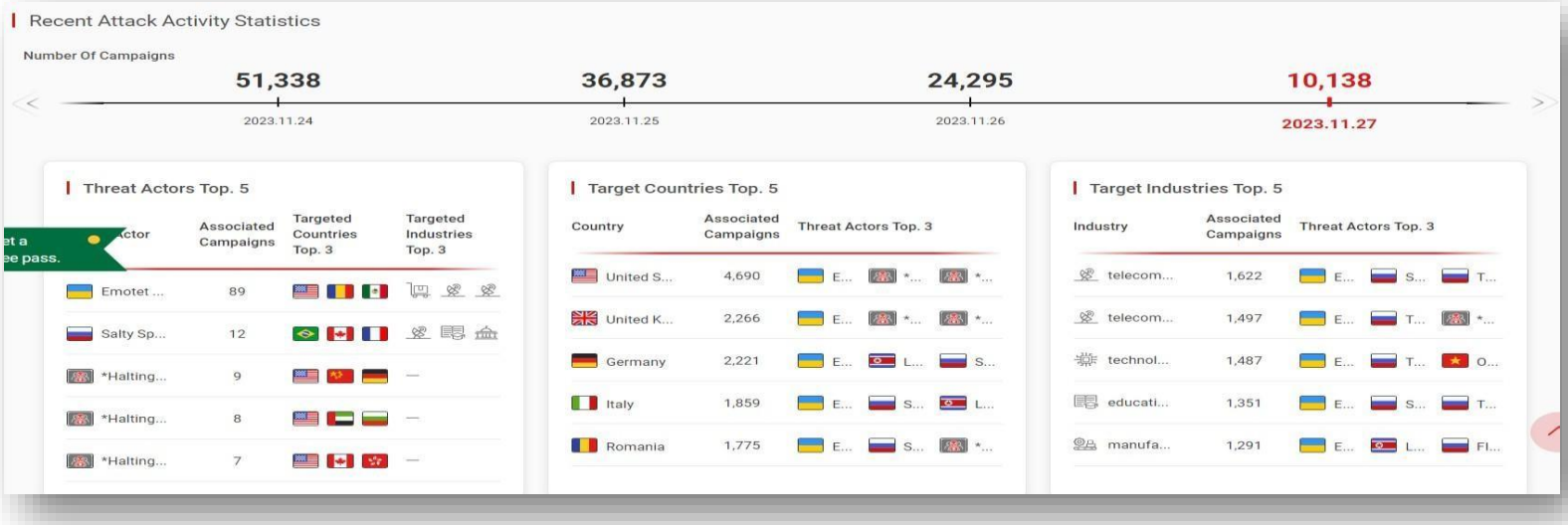
Skill stack

FE: Javascript-Vue.js
BE: Java-Spring Boot, MySQL

RNRs

Front-End





Development of Commercial Web ㉮

Freelancer

PJT Outline

- Developed Commercial Webpage *hansolmeat.shop*
- Published all FE/BE/DB part

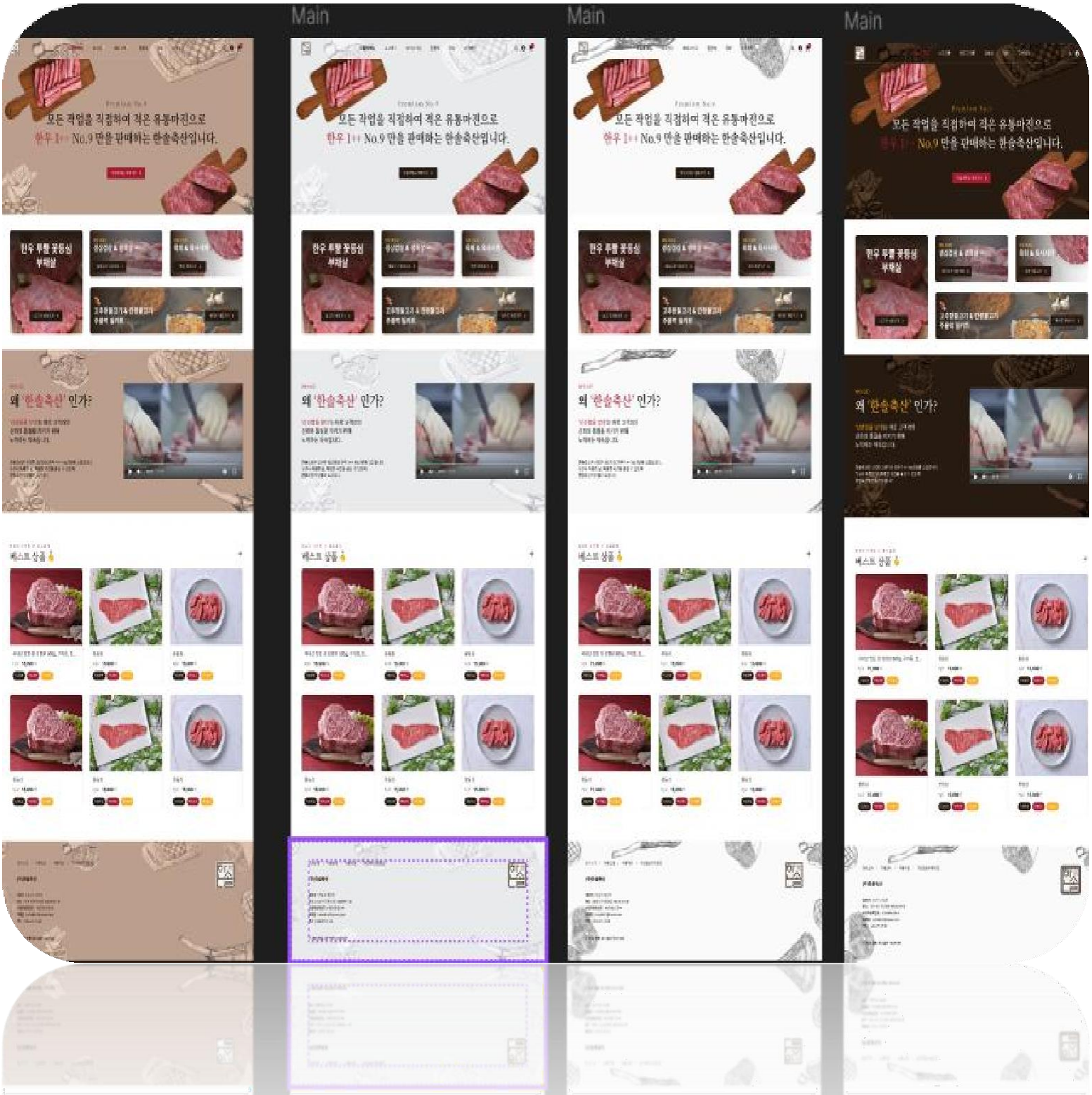
Skill stack

FE: Next.js

BE: Next.js, MySQL, MariaDB

RNRs

PM, PL



Development of Commercial Web ㉯

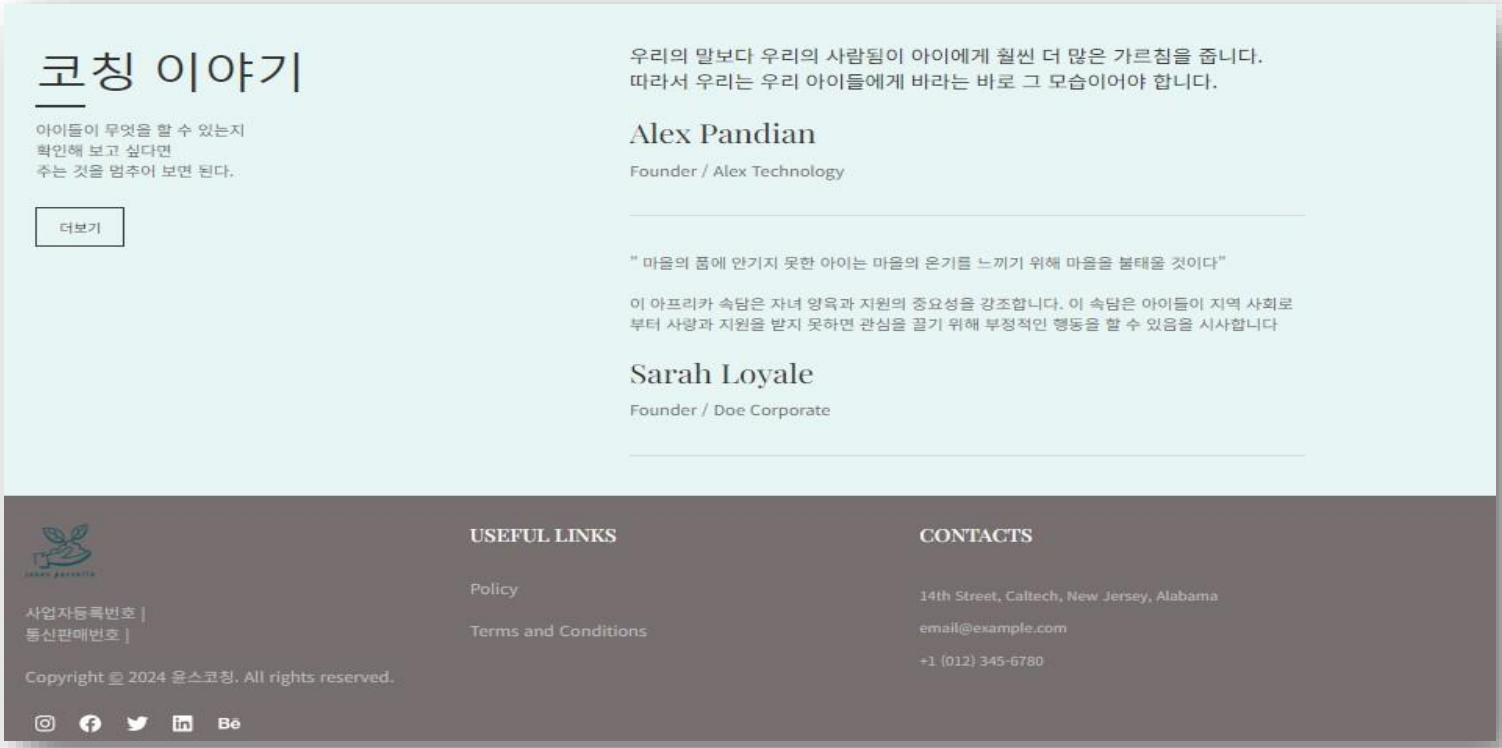
Freelancer

PJT Outline

- Developed Commercial Webpage *yoonsparenting.com*

Skill stack

- Wordpress

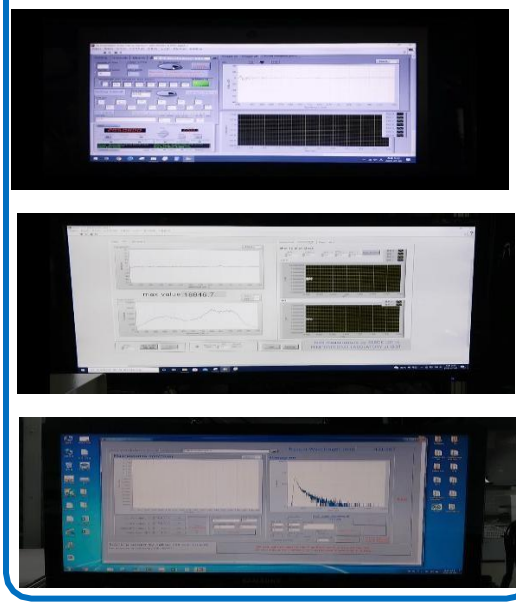


Metal-Enhanced Fluorescence

Femtosecond Spectroscopy Lab. Advisor: Dr.Yoonsoo Pang
Gwangju Institute of Science and Technology

Investigated MEF phenomena of DCM & C343 dyes near Ag NPs Analyzed data from TCSPC laser

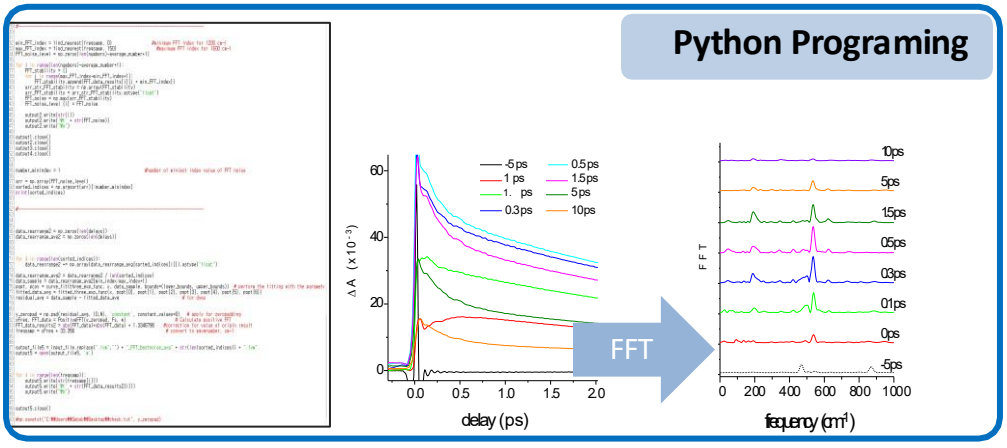
LabVIEW Programing



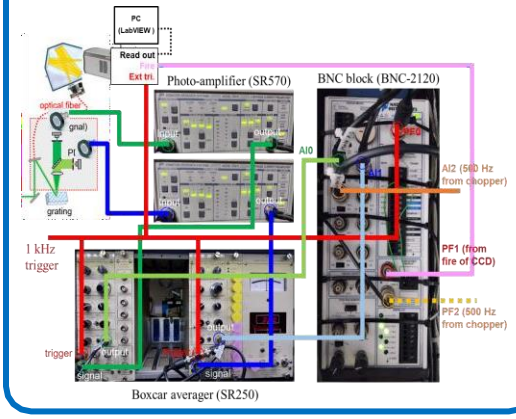
Development of femtosecond time-resolved spectrometer system

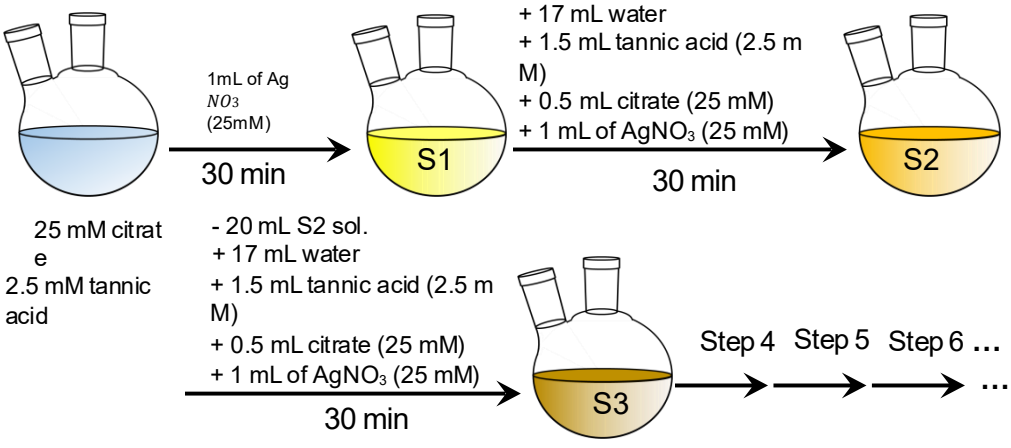
LabVIEW & Python Programing

Python Programing



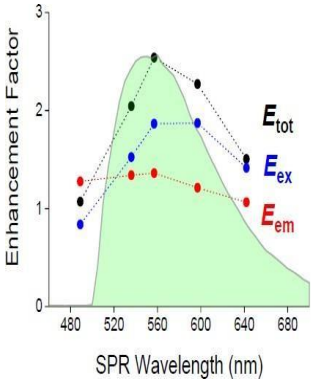
H/W system design





Quantum Yields of DCM & C343 probed at 405nm in coated thin PS films with SCFs

	Lifetime (ns)	Quantum Yield, Q_m	Emission Enhancement Factor, E_{em}	Total Enhancement Factor, E_{tot}	Excitation Enhancement Factor, E_{ex}
DCM Only	1.59	0.47	-	-	-
DCM with SCF1	1.19	0.60	1.28	1.07	0.84
DCM with SCF2	1.11	0.63	1.34	2.04	1.53
DCM with SCF3	1.09	0.64	1.36	2.54	1.87
DCM with SCF4	1.30	0.57	1.21	2.27	1.87
DCM with SCF5	1.50	0.50	1.06	1.51	1.41



$$Q_0 = \frac{\Gamma_0}{\Gamma_0 + k_{nr}} = \tau_0 \Gamma_0$$

$$Q_m = \frac{\Gamma_m}{\Gamma_m + k_{nr,m}} = \tau_m \Gamma_m$$

$$E_{em} = \frac{Q_m}{Q_0}$$

Q_0 of DCM in PS = 0.47

Semi-empirical model

$$Q_0 = \frac{\Gamma_0}{\Gamma_0 + k_{nr}} = \tau_0 \Gamma_0$$

$$Q_m = \frac{\Gamma_m}{\Gamma_m + k_{nr}} = \tau_m \Gamma_m$$

with SCFs

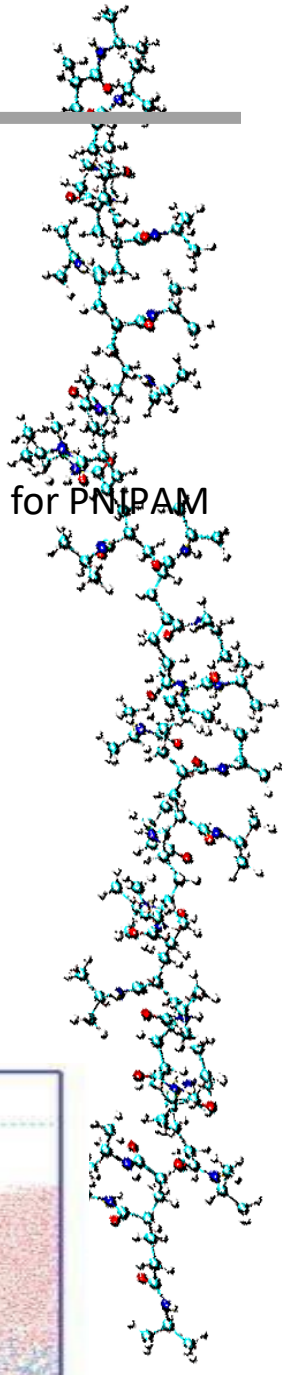
$$\Gamma_m = \frac{1}{\tau_m} - k_{nr} = \frac{1}{\tau_m} - \frac{(1 - Q_0)}{\tau_0}$$

$$Q_m = 1 - \frac{\tau_m}{\tau_0}(1 - Q_0)$$

$$E_{em} = \frac{Q_m}{Q_0} \qquad E_{ex} = \frac{E_{tot}}{E_{em}}$$

Molecular Dynamics Simulation of PNIPAM

Computational Polymer Physical Chemistry Lab. Advisor: Dr.Jun Soo Kim
Ewha Womans University



Skill stack

Python, Fortran, CentOs, Gaussian05

PJT Outline

-MD Simulation
-Data Analysis using FORTRAN

Force field: OPLS-AA
Model: Tip4p/2005 for water
Syndiotactic arrangement of side groups made by Gauss View 5.0 program for PNIPAM

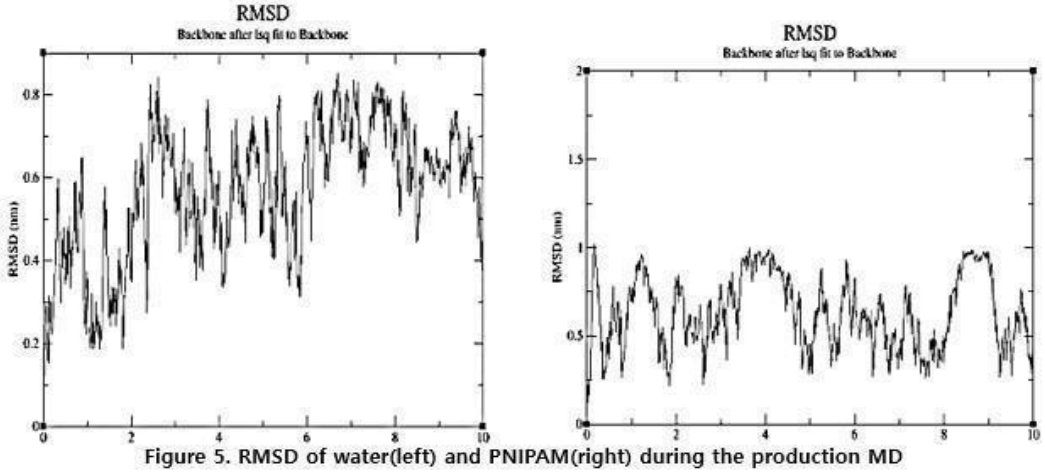
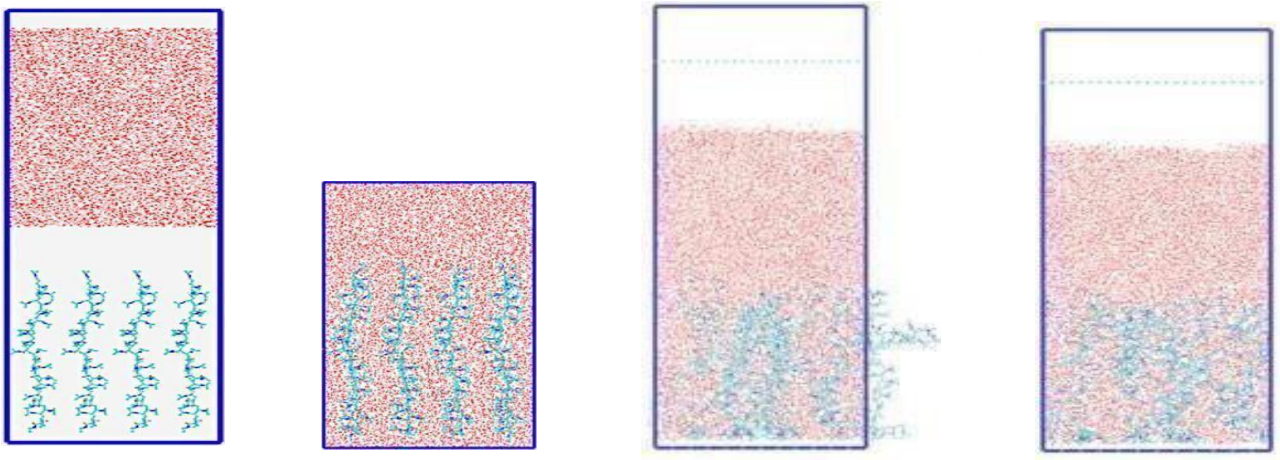


Figure 5. RMSD of water(left) and PNIPAM(right) during the production MD



- Developed PNIPAM NXN multi-chains system and conducted MD simulations
- Analyzed Radius of gyration and intra-chain distances to elucidate collapse-swelling phase transitions
- To advance the research further, development of coarse grained parameters are necessary with reduced degrees of freedom by overcoming time-scale limitations
- Follow-up Study: Interaction of PNIPAM molecules with DNA

Face-Age Detector

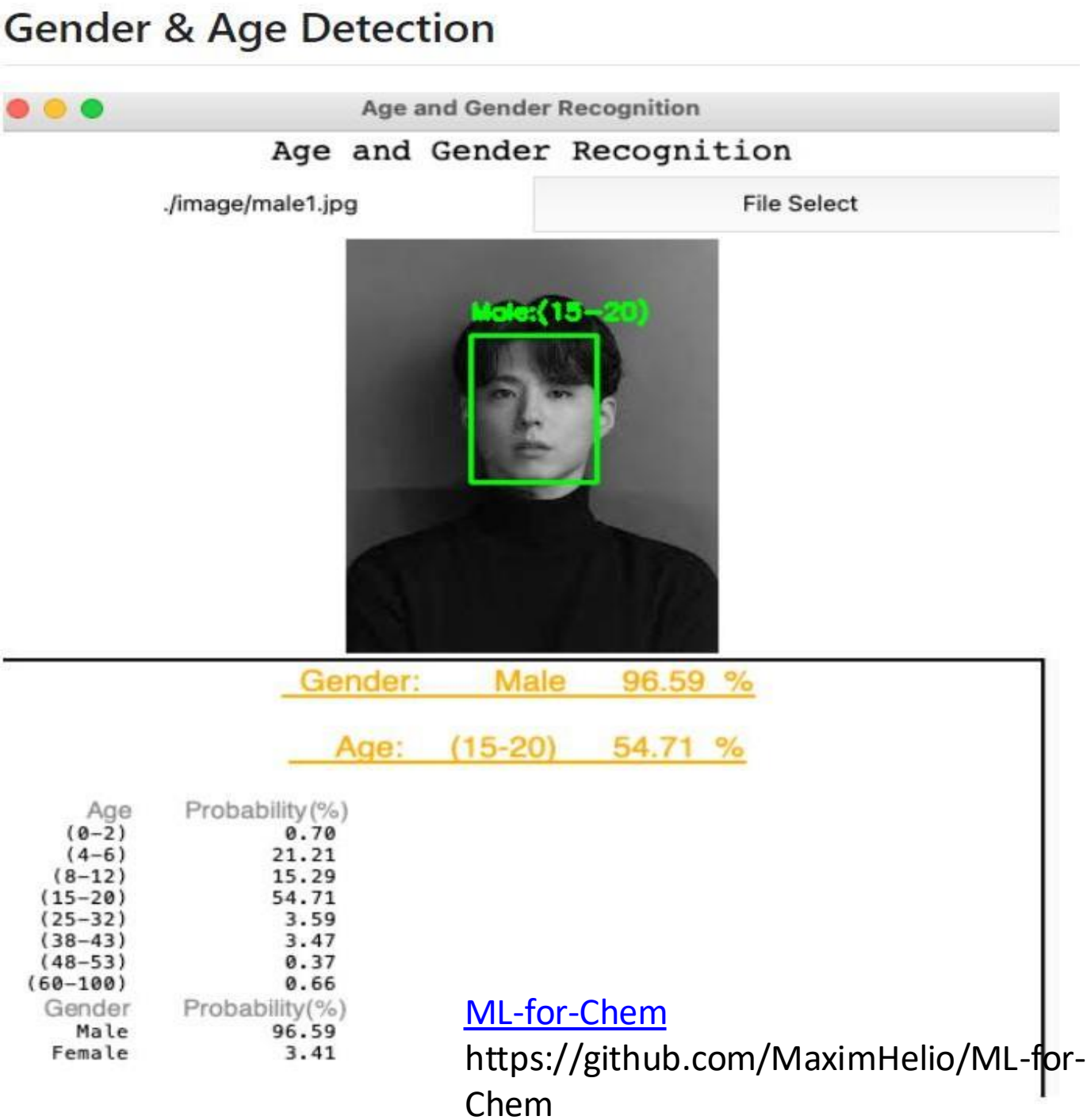
Personal PJT

PJT Outline

- Developed face-age detector program using Caffemodel

Skill stack

DNN, caffemodel, PIL, Caffe framework



[ML-for-Chem](https://github.com/MaximHelio/ML-for-Chem)

https://github.com/MaximHelio/ML-for-Chem

BOJ Study Batch Alarm

https://github.com/MaximHelio/BOJ-cote-batch

Personal PJT

Skill stack

FE: React.js

BE: SpringBoot JPA, Spring Batch

PJT Outline

Every day (MON-SUN) at 00:10 Check the number of problems solved by BOJ User

1. Run SBApplication
2. Enter the following into the kknni_tb_user table in MySQL client

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

id	email	joinType	name	password
1	duruaal@naver.com	일반	duruaal	123456
2	sgs1159	일반	sgs1159	123456
NULL	NULL	NULL	NULL	NULL

3. Run at localhost:9040/user/rank > result

#Algorithm

Participants: Sgs1159,Duruaal

No	Date	Title	Baekjoon
1	2023-02-19	Duruaal	✗ (1)
2	2023-02-19	Sgs1159	✗ (0)
3	2023-02-19	Duruaal	✗ (1)
4	2023-02-19	Sgs1159	✗ (0)
5	2023-02-19	Duruaal	✗ (1)
6	2023-02-19	Sgs1159	✗ (0)

Github | Instagram

Blog Automatic Writing Bot

https://github.com/MaximHelio/btw_newedit

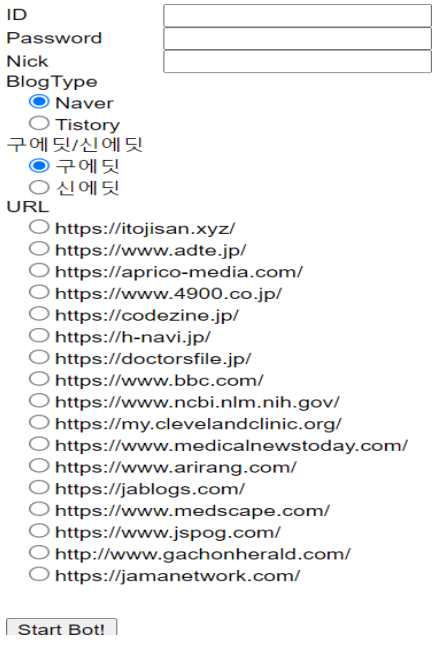
Personal PJT

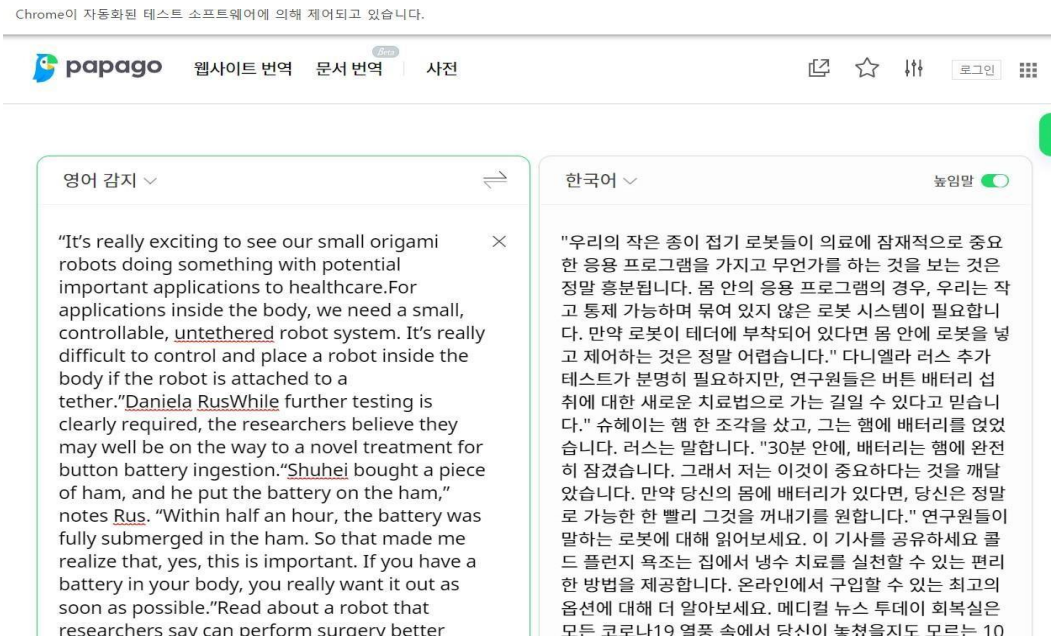
Skill stack

Node.js
Jade

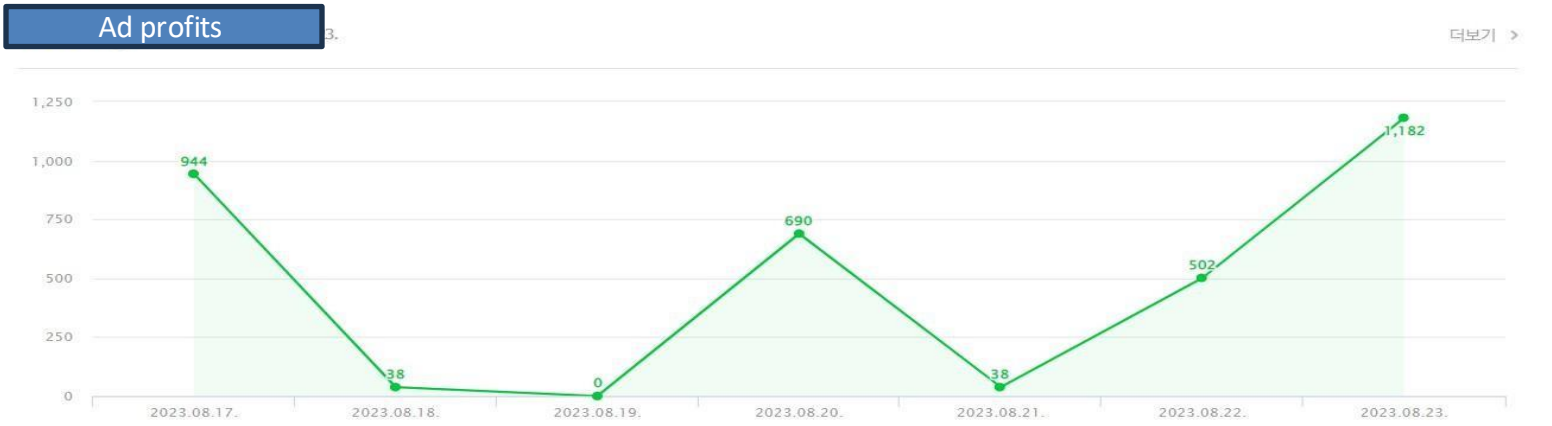
PJT Outline

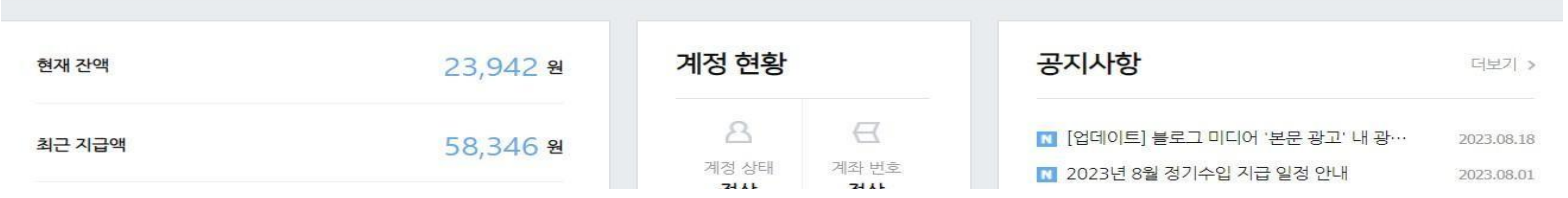
A bot that automatically writes to a blog when run.
Achieved profits.





Ad profits





Rating-based Movie Community Service Web

https://github.com/MaximHelio/SSAFY-Final-pjt-SGMovies

Personal PJT

Skill stack

FE: Vue.js

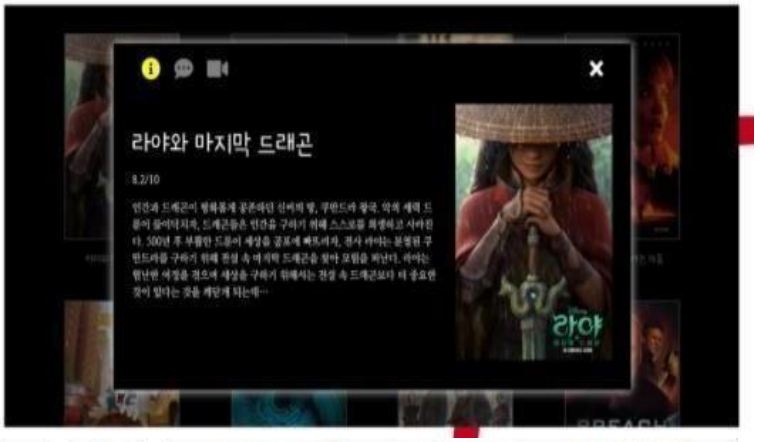
BE: Django

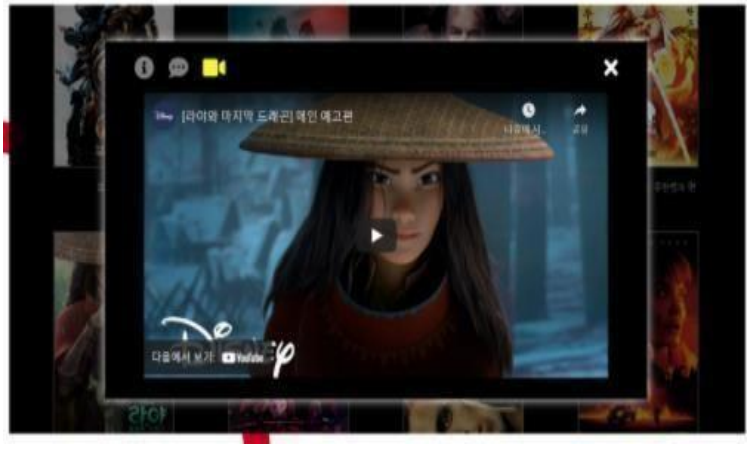
YouTube API, TMDB API, JWT, Sqlite3

PJT Outline

Developed a movie information bulletin board, including comments and recommendation functions

Implemented Search algorithms for content corresponding to keywords users inserted.





- JWT Login
- Applied YouTube API
- Recommendation system